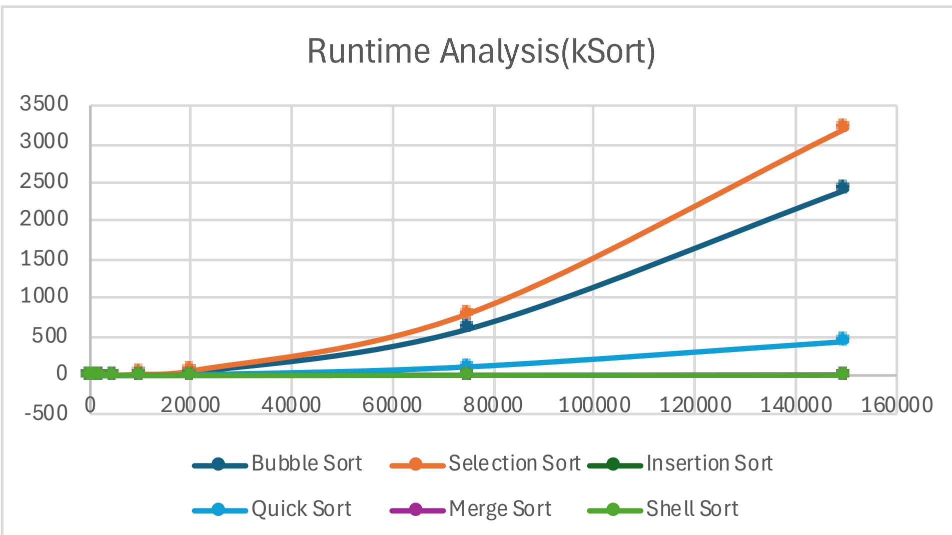
Sizes	Bubble Sort	Selection Sort	Insertion Sort	Quick Sort	Merge Sort	Shell Sort
100	0	0	0	0	0	0
500	0	0	0	0	0	0
1000	0	0	0	0	0	0
2000	0	0	0	0	0	0
5000	2	3	0	0	0	0
10000	7.25	14	0	2	0	0
20000	37.2	56.85	0	7.9	0.75	0
75000	601.3	801.05	1.8	111.1	3.65	1.05
150000	2407.25	3204	5.4	439.15	6.5	3.05

The performance of the k-sorted data is pretty similar to the preformance of the non k-sorted data. The only major difference is the preformance of selection sort and quick sort. Quick sort preforms better in randomized arrays because the pivots that it selects will be more randomized. In k-sorted array the pivot point is more likely to be high or to low to sort the array effeciently. Selection sort preforms worse with k-sorted arrays because it will still have to scan through the rest of the array everytime. So even if the array is more sorted, it



Runtime Analysis(kSort)
3500
3000
2500
2000
1500
1000
500
0
-500
0
20000
40000
60000
80000
100000
120000
140000
160000
Bubble Sort
Selection Sort
Insertion Sort
Quick Sort
Merge Sort
Shell Sort